Lab 3: Architectural Justification

Hospital Bed & ICU Allocation Dashboard

Architecture Selection: We chose a Client-Server Architecture for the Hospital Bed & ICU Allocation Dashboard.

Reason 1 — A single, authoritative source of bed-state truth

Bed occupancy and ICU allocation decisions directly affect patient safety, so two dashboards must never disagree about whether a bed is free. A centralized Bed Allocation Server acting as the single source of truth avoids the eventual-consistency and data-synchronization issues a distributed microservices split would introduce across ER dashboard clients.

Reason 2 — Natural fit for real-time broadcast to many clients

NFR-001 requires bed status transitions to propagate to all connected ER dashboard clients in under one second via WebSockets. A centralized server holding live state can push updates to every connected Triage Nurse or Hospital Administrator client simultaneously and consistently, which is exactly what the client-server broadcast model is built for.

Security Advantage

Patient vitals, triage scores, and bed/ICU status are sensitive healthcare data. Centralizing all access through the Bed Allocation Server means authentication, authorization, and audit logging for every read or update happen at one enforced boundary, rather than being duplicated (and potentially weakened) across many independently deployed services.

Performance Benefit

Because the WebSocket Broadcast Service sits close to the server's in-memory bed state and pushes updates directly rather than clients polling or coordinating across distributed nodes, the system can reliably meet the sub-second propagation target even during peak ER load.

Component Diagram

